HEIGHTS AND TORSION ON ELLIPTIC CURVES

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Kronecker's theorem says that the elements of $\overline{\mathbb{Q}}^\times$ of height 1 are the roots of unity, the torsion of $\mathbb{G}_\mathrm{m}$. This worksheet proves its analogue for an elliptic curve, for the height h_x of the x-coordinate, assuming the duplication bound of [Sil09, Corollary VIII.6.4(b)]. The torsion points have bounded height, not height 0, and that is enough for finiteness. First come the valuations of a sum of several terms, and the shape of the coordinates of a rational point, over $\mathbb{Q}$ and over a number field: the first step of Silverman's proof of the Mordell–Weil theorem over $\mathbb{Q}$. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A number field k, elements $a, b \in k$ with $\Delta := -16(4a^3 + 27b^2) \neq 0$, and the elliptic curve $E_{a,b}\colon y^2 = x^3 + ax + b$ over k.
- The height H on $\mathbb{P}^N(\overline{\mathbb{Q}})$, and $h := \log H$. For $t \in \mathbb{Q}$, with $t = u/w$ in lowest terms, $H([t:1]) = \max(|u|, |w|)$. See [Sil09, §VIII.5].
- The x-coordinate $x\colon E_{a,b} \to \mathbb{P}^1$, with $x(O) = [1:0]$, and the height $h_x(P) := h(x(P))$ of a point $P \in E_{a,b}(\overline{\mathbb{Q}})$. So $h_x(O) = 0$.

How to use this. Allow about 10 minutes for Problem 1.2, 15 for Problem 2.2 and 20 for Problem 3.4.

1. Valuations of sums

The proofs below compare the valuations of the terms of a sum. This section makes the ultrametric inequality precise: when it is an equality, and what happens when it is not.

Definition 1.1 (Discrete valuations)**.** A discrete valuation on a field F is a surjective map $v\colon F \to \mathbb{Z} \cup \{\infty\}$ with $v(\alpha) = \infty$ only for $\alpha = 0$, $v(\alpha\beta) = v(\alpha) + v(\beta)$ and $v(\alpha + \beta) \geqslant \min(v(\alpha), v(\beta))$. Its valuation ring is $\mathcal{O}_v := \{\alpha \in F : v(\alpha) \geqslant 0\}$, a uniformizer is an element $\pi \in F$ with $v(\pi) = 1$, and the residue field is $\kappa := \mathcal{O}_v/\pi\mathcal{O}_v$, with reduction $\alpha \mapsto \bar{\alpha}$. Examples are v_p on $\mathbb{Q}$, and $v_\mathfrak{p}$ on a number field k, for a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$.

Problem 1.2 (The unique minimizer criterion)**.**
Let v be a discrete valuation on a field F, with uniformizer π and residue field κ. Let $\alpha_1, \ldots, \alpha_n \in F$, not all 0, $m := \min_i v(\alpha_i)$, and $I := \{i : v(\alpha_i) = m\}$.
(1) Show that $v(\alpha_1 + \cdots + \alpha_n) \geqslant m$, with equality if I has one element.
(2) Show that $v(\alpha_1 + \cdots + \alpha_n) = m$ if and only if $\sum_{i \in I} \overline{\alpha_i/\pi^m} \neq 0$ in κ. Give an example with $\#I = 2$ where the inequality is strict.

Date: November 5, 2026.

2. The shape of a rational point

This section is the first step of the proof of [Sil09, Lemma VIII.4.2], which proves that h_x satisfies the hypotheses of the descent theorem over $\mathbb{Q}$. It turns the rational points of bounded height into a finite search over integers. The second part is the same argument over a number field, where ideals take the place of the integer d. The integrality of a, b is harmless: if u^4a, u^6b are integral, then $(x, y) \mapsto (u^2x, u^3y)$ is an isomorphism onto E_{u^4a,u^6b}, and it changes h_x by at most $2\log|u|$ for $u \in \mathbb{Z}$.

Definition 2.1 (Heights over a number field [Sil09, §VIII.5])**.** For $t \in k$,

$$H([t:1])^{[k:\mathbb{Q}]} = \prod_{v \in \mathcal{P}_k} \max(1, |t|_v)^{n_v},$$

with $n_v \coloneqq [k_v : \mathbb{Q}_p]$ or $[k_v : \mathbb{R}]$. For the place of a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$, $|t|_v^{n_v} = N(\mathfrak{p})^{-v_\mathfrak{p}(t)}$, where $N(\mathfrak{p})$ is the order of $\mathcal{O}_k/\mathfrak{p}$. The denominator ideal of t is $\mathfrak{d}(t) \coloneqq \prod_\mathfrak{p} \mathfrak{p}^{\max(0,-v_\mathfrak{p}(t))}$, an ideal of $\mathcal{O}_k$.

Problem 2.2 (The shape of a rational point)**.**
(1) Let $a, b \in \mathbb{Z}$, and $P \in E_{a,b}(\mathbb{Q}) \smallsetminus \{O\}$. Show that $x(P) = n/d^2$ and $y(P) = m/d^3$ with $n, m, d \in \mathbb{Z}$, $d \geqslant 1$ and $\gcd(n, d) = \gcd(m, d) = 1$, and that $H(x(P)) = \max(|n|, d^2)$.[a]
(2) Let $a, b \in \mathcal{O}_k$, and $P \in E_{a,b}(k) \smallsetminus \{O\}$. Show that there is an ideal $\mathfrak{d}$ of $\mathcal{O}_k$ with $\mathfrak{d}(x(P)) = \mathfrak{d}^2$ and $\mathfrak{d}(y(P)) = \mathfrak{d}^3$, and that

$$H(x(P))^{[k:\mathbb{Q}]} = N(\mathfrak{d})^2 \prod_{v|\infty} \max(1, |x(P)|_v)^{n_v}.$$

Check that for $k = \mathbb{Q}$ this is (1).

[a] *Hint:* compare $v_p(x)$ and $v_p(y)$ at a prime p with $v_p(x) < 0$.

3. Kronecker's theorem for elliptic curves

Kronecker's theorem [Sil09, Exercise 8.6] says that $\alpha \in \overline{\mathbb{Q}}^\times$ has $H(\alpha) = 1$ if and only if α is a root of unity. This section proves the analogue for $E_{a,b}$ and h_x, from the results below.

Proposition 3.1 (The duplication formula [Sil09, Algorithm III.2.3(d)])**.** *Let $P \in E_{a,b}(\overline{\mathbb{Q}})$ with $2P \neq O$, and $x \coloneqq x(P)$. Then*

$$x(2P) = \frac{x^4 - 2ax^2 - 8bx + a^2}{4(x^3 + ax + b)}.$$

Proposition 3.2 (The duplication bound [Sil09, Corollary VIII.6.4(b)])**.** *There is a constant $C \geqslant 0$, depending on a and b, with $|h_x(2P) - 4h_x(P)| \leqslant C$ for all $P \in E_{a,b}(\overline{\mathbb{Q}})$.*

Theorem 3.3 (Northcott [Sil09, Theorem VIII.5.11])**.** *For all constants B and D, the set of $R \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(R) \leqslant B$ and $[\mathbb{Q}(R) : \mathbb{Q}] \leqslant D$ is finite. Here $\mathbb{Q}(R)$ is the field of definition of R.*

Problem 3.4 (Kronecker for elliptic curves)**.**
Let C be the constant of Proposition 3.2. Only its lower bound, $h_x(2P) \geqslant 4h_x(P) - C$, is used.
(1) Show that every torsion point $T \in E_{a,b}(\overline{\mathbb{Q}})$ has $h_x(T) \leqslant C/3$.[a]
(2) Show that for every $D \geqslant 1$ there are only finitely many torsion points $T \in E_{a,b}(\overline{\mathbb{Q}})$ with $[\mathbb{Q}(T) : \mathbb{Q}] \leqslant D$. Deduce that the torsion subgroup of $E_{a,b}(K)$ is finite for every number field K containing k.

(3) Show that the naive analogue of Kronecker's theorem fails in both directions: a torsion point need not have $h_x = 0$, and a rational point with $h_x = 0$ need not be a torsion point.

(a) On $y^2 = x^3 + 1$, show that $T = (2, 3)$ is a torsion point of order 6, with $h_x(T) = \log 2$.

(b) On $y^2 = x^3 + x + 1$, the rational point $P = (0, 1)$ has $h_x(P) = 0$. For this curve over $\mathbb{Q}$, one can take $C = \log 619$ in the lower bound $h_x(2P) \geqslant 4h_x(P) - C$. Show that P has infinite order.[b]

[a] *Hint:* the multiples of T form a finite set that contains the double of each of its elements. Look at the one of largest height.

[b] *Hint:* compute $x(2P)$ and $x(4P)$.

References

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑